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Thesis/Project title: Design of a Multi-Channel MPPT System for Solar Vehicle Applications

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**SCHOOL OF ELECTRICAL ENGINEERING
AND TELECOMMUNICATIONS**

Design of a Multi-Channel MPPT System for Solar Vehicle Applications

by

Angus Macdonald

Student ID: z5416730

Thesis submitted as a requirement for the degree
Bachelor of Engineering (Electrical Engineering)

Submitted: April 23, 2026
Supervisor: Prof. John Fletcher

Abstract

Sunswift Racing is an ambitious student project focused on pushing the boundary of solar car technology. In the past, their developments have focused on solar race cars for the BWSC competition. The team has now transitioned to developing their next vehicle, Sunswift 8. This is a *TriBrid* ADR compliant vehicle that contains three distinct power sources, battery, hydrogen, and solar power. Integration of solar arrays into automobiles is a relatively niche market given the engineering and manufacturing complexity needed to derive useful amounts of energy from the small usable surface area on a traditional car. Efficient solar extraction for automobiles comes with a unique set of challenges that differ greatly from conventional solar projects. These include, but are not limited to; large differences in effective irradiance between arrays, frequent partial shading, varying temperature gradients, tight space constraints, and harsh automotive environments. The goal of this thesis is to develop an integrable product for Sunswift 8 that extracts maximal energy from the onboard photovoltaic arrays and delivers it to the LV and HV batteries in the car. Traditionally, solar arrays are connected in series or parallel and a single MPPT is used to extract power. This thesis proposes per-string power-point tracking using synchronous boost converters, with all MPPT modules integrated on a single board. It will also act as a robust system to deliver power to the LV bus and incorporate a separate power stage to allow HV EV battery charging.

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Abbreviations

PV Photovoltaic

ADR Australian Design Rules

BWSC Bridgestone World Solar Challenge

TriBrid An automobile with three energy sources

SR8 Sunswift 8

LV Low voltage

HV High voltage

MPP Maximum power point

MPPT Maximum power point tracker

SMPPT Single maximum power point tracker

EV Electric vehicle

SC Short circuit

OC Open circuit

DC Direct current

PWM Pulse width modulation

CAN Controller area network

COTS Commercial Off-The-Shelf

BMS Battery management system

STC Standard test conditions

HJT Heterojunction

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Chapter 1

Introduction

Photovoltaic (PV) technology is a widely adopted method of electrical power generation due to its ability to convert light into electrical energy[8]. This has enabled integration across a broad range of industries, including aerospace, grid-scale power generation, and embedded systems. However, the integration of PV arrays into automotive platforms introduces a distinct set of engineering constraints that are not present in conventional PV systems[3]. Limited surface area, non-uniform irradiance, and dynamic operating conditions lead to significant mismatch losses in PV systems[12]. As a result, effective utilisation of PV technology in automotive applications requires specialised system design to address these challenges[1].

1.1 Background

PV cells exhibit a non-linear IV characteristic, meaning that a maximum power point tracker (MPPT) must be used to extract power at the correct operating point for maximum efficiency. The IV characteristic curve of a PV cell is impacted by environmental factors which change the maximum power point (MPP), so the control algorithm must allow the MPPT to constantly track and recognise this MPP. The temperature, irradiance spectrum shape, partial shading, effective irradiance angle, cell age, and irradiance level all shift the MPP of the panel non-linearly as this affects the I_{MPP} and V_{MPP} of an individual cell depending on the environmental factor.

PV arrays are comprised of PV cells, and these cells can be configured in series or parallel. Often arrays are configured in series strings, and multiple strings may be configured in parallel to be connected to a single MPPT. A major problem arises with this approach, in the form of

mismatch losses. Mismatch losses occur when one cell experiences different environmental factors to other cells in a string. This causes the I_{MPP} of this cell to become smaller than other cells in the string, limiting the other cells of the string to the I_{MPP} of the weakest cell, creating major efficiency losses.

There are many ways to configure PV arrays and MPPT placement to optimise PV technology for a specific use case, and to help negate mismatch losses that lower the efficiency of PV arrays. The pronounced mismatch losses in automotive systems renders SMPPT solutions impractical for efficient operation.

1.2 Problem Statement

Environmental factors are present in any PV system, and precautions must be taken to ensure that PV strings do not become affected by mismatch losses. The most prominent causes of mismatch losses are due to effective irradiance angles and partial shading. In conventional solar projects, solar panels are designed to be flat to minimise differences in irradiance angle, and solar panels are placed such that partial shading affects do not occur. For example, a solar farm is placed in a wide open area with no trees or obstructions to cause shading, and the irradiance angle is consistent in each panel due to the flat panel design.

In the design of SR8, these luxuries could not be afforded. PV arrays could only be placed on the curved aerodynamic body panels, and these arrays cannot be placed on the same plane. This means that the arrays cannot be connected in series with a SMPPT, otherwise major mismatch losses will occur.

An automotive environment also exhibits highly dynamic operating conditions dissimilar to any other PV system environment creating large mismatch losses. This means that a more complex MPPT system is required to efficiently extract usable amounts of power from PV arrays on a vehicle. Given the pronounced mismatch effects due to panel angle and environmental conditions, a multi-channel MPPT architecture is required.

1.3 Thesis Objectives

This thesis is focused on the design of a MPPT module for automobiles with integrated PV arrays that allows for maximal efficiency in the highly dynamic environmental conditions that affect PV arrays on vehicles. This MPPT module will be developed in conjunction with Sunswift Racing, with design decisions focused on optimising the power extraction from SR8's onboard PV arrays.

The ultimate objective is to deliver Sunswift Racing a robust integrable product that requires very little configuration to receive optimal performance. This product will acknowledge the unique operating conditions of a PV system in an automotive environment, and take appropriate design measures to limit or negate mismatch losses in this system. This product can be broken down into four main objectives which are outlined below:

1. Design and validate a per-string MPPT architecture
2. Develop and validate a HV conversion stage capable of interfacing LV MPPT outputs with the HV EV battery system.
3. Research and integrate existing MPP tracking algorithms into this system. Develop the firmware required to safely integrate this MPPT system into the SR8 electrical architecture.
4. Integrate LV, HV, and firmware design components into a single product that can be deployed in the SR8 vehicle.

1.4 Chapter Outline

This report is written with 5 main sections. Chapter 2 reviews the background surrounding MPPT systems and PV systems, along with existing solutions for efficiently extracting power from PV arrays in automotive systems. Chapter 3 overviews the proposed solution to this problem in detail. Chapter 4 outlines work undertaken in thesis A to progress the thesis objectives. Chapter 5 is a summary of the plan to complete the thesis objectives and provide Sunswift Racing with this product by the end of this year. Chapter 6 summarises the key aspects of this report.

Chapter 2

Literature Review

2.1 Photovoltaic Cell Characteristics

Photovoltaic (PV) cells convert the energy in light to electrical power by using the photovoltaic effect[8]. Photons impact a semiconductor material and displace electrons, creating electron-hole pairs and inducing a current. This generates a DC voltage across the PV cell. PV cells can be arranged in series or parallel like a battery, increasing terminal voltage or current, respectively. PV cells exhibit a non-linear IV curve[17]. This is illustrated in figure 2.1.

2.1.1 The IV Curve

Understanding the behaviour of a PV IV curve is critical to designing and creating a MPPT. Electrical components such as resistors have linear IV characteristics, which means that $I \propto V$. The IV characteristic of a PV cell can be described as a "bent" shape. As the voltage increases from zero, the current remains unchanged. As you continue to increase the voltage past a certain threshold, the current will drop to zero. The IV curve can be translated into a PV curve by calculating $P = IV$ at every point. This clearly describes where the MPP of the PV cell occurs.

All commercial PV cells describe this behaviour with 4 critical parameters.

V_{OC} The open-circuit (OC) voltage occurs at $I = 0$, when there is no load on the cell.

I_{SC} The short-circuit (SC) current occurs at $V = 0$, when there is a short across the cell.

V_{MPP} The maximum power point (MPP) voltage is an intrinsic parameter determined by PV

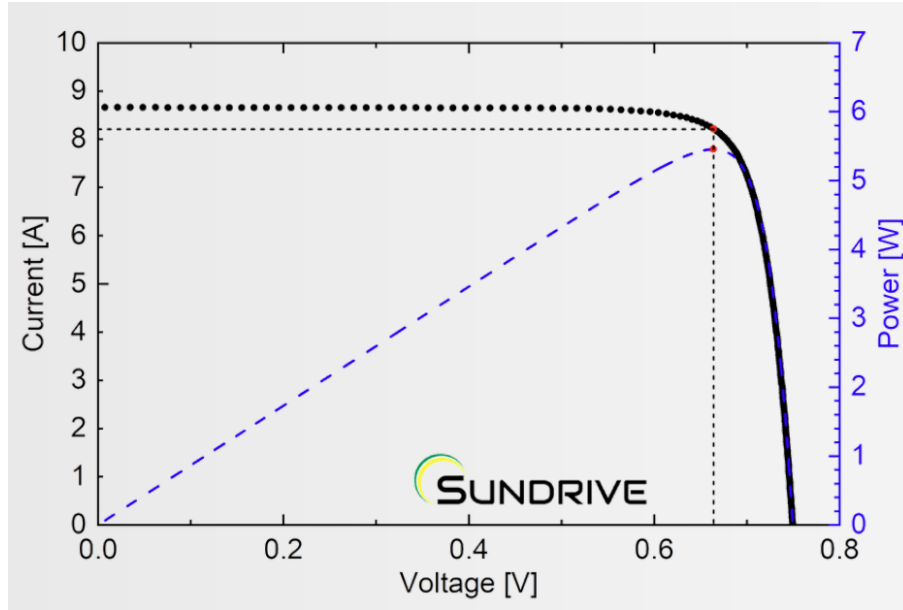


Figure 2.1: PV Curve of Sundrive G12-210 SD-HJT cell[15]

cell construction. This is the desired operating voltage of the panel for maximal efficiency.

I_{MPP} The maximum power point (MPP) current is an intrinsic parameter determined by PV cell construction. This is the desired operating current of the panel for maximal efficiency.

Commercial solar cells measure these key parameters under standard test conditions (STC). These conditions are 1000 W/m^2 of irradiance, 25°C cell temperature, and an AM1.5 air mass spectrum.

These parameters are not fixed and are highly dynamic based on PV cell inputs and environmental factors. This means that PV cells cannot be loaded simply at the STC MPP operating point, and the shape and position of the IV and PV curves are constantly changing. This is the reason PV systems require an MPPT for efficient operation[5].

2.1.2 Impact of Irradiance on the IV Curve

The core mechanism that changes the shape of the IV curve is the irradiance (G) received by the PV cell. Irradiance is an indicator of the energy in light. A higher irradiance indicates a greater photon flux incident on the cell surface. At a high level, this displaces more electrons in the PV cell semiconductor material, generating more electron-hole pairs, and therefore increasing the

photocurrent in the PV cell. This effect is reflected in the IV curve, as $I_{SC} \propto G$ [8]. This increases I_{MPP} approximately linearly and, therefore, P_{MPP} . PV cells also exhibit diode characteristics that govern the voltage response to changes in irradiance, encapsulated in equation 2.1. When $I = 0$, the equation can be rearranged as equation 2.2.

$$I = I_{ph} - I_0 \left(e^{\frac{V}{nV_T}} - 1 \right) \quad [17] \quad (2.1)$$

$$V_{OC} = nV_T \ln \left(\frac{I_{ph}}{I_0} + 1 \right) \quad [17] \quad (2.2)$$

Where I_{ph} is the photocurrent generated by incident irradiance, I_0 is the diode saturation current, n is the diode ideality factor, and V_T is the thermal voltage given by $V_T = \frac{kT}{q}$.

Since I_{ph} is approximately proportional to irradiance, equation 2.2 shows that V_{OC} increases logarithmically with G [17]. As a result, changes in irradiance produce only a small variation in voltage compared to the linear increase in current.

The irradiance received by the PV cell surface is dependent on the angle of incidence. Equation 2.3 shows the effective irradiance received by a cell when the cell is oriented at an angle θ to the incoming light.

$$G_{eff} = G \cos(\theta) \quad [8] \quad (2.3)$$

In PV systems like SR8 where cell arrangements are curved or multiple arrays are oriented differently, the result is a difference in irradiance received by the cells. This means that PV cells will have different IV curves and MPP operating points. If PV cells are connected in series, the string current will be constrained by the cell generating the lowest photocurrent, resulting in a reduced P_{MPP} .

2.1.3 Impact of Temperature on the IV Curve

Temperature also affects the shape of the IV curve. PV cell data sheets typically provide temperature coefficients describing the variation of P_{MPP} , V_{OC} , and I_{SC} with temperature. An increase in temperature results in a slight increase in I_{SC} due to enhanced carrier generation[8]. However, the diode saturation current I_0 increases exponentially with temperature, while the thermal voltage V_T increases linearly, as seen in equation 2.4[17]. From Equation 2.2, the exponential increase in I_0 dominates, resulting in a significant reduction in V_{OC} [17]. Consequently, the IV curve shifts to lower voltages, leading to a reduction in V_{MPP} and the overall

power output[8].

$$V_T = \frac{kT}{q} \quad (2.4)$$

Other factors such as cell degradation, partial shading, and non-uniform irradiance spectrum also influence the IV curve. In particular, non-uniform conditions across PV arrays introduce significant deviations in operating behaviour between PV cells. This is examined in detail in the following section.

2.2 Partial Shading and Mismatch Losses

Commercial PV cells are modular by design and are typically interconnected to form application specific configurations. PV cells are connected in series to form strings, and multiple strings may be connected in parallel to form an array. In series connected strings, the same current must flow through all cells, which constrains performance under non-uniform conditions[7]. When strings are connected in parallel, the terminal voltage of all strings must be equal[7]. Parallel mismatch is less severe than series mismatch, but can still reduce total power.

Mismatch occurs when PV cells exhibit different IV characteristics due to variations in irradiance, temperature, or orientation[7]. As a result, each cell has distinct V_{MPP} and I_{MPP} values. Under uniform conditions, cells have similar IV curves and can operate near a common operating point. However, under mismatch conditions, the interconnected system must operate at a single operating point, preventing each cell from operating at its individual MPP[7].

In series-connected strings, this effect is particularly pronounced, as the same current must flow through all cells. The string current is therefore limited by the cell producing the lowest current, forcing higher-performing cells to operate away from their optimal point[12]. This results in mismatch losses, defined as the difference between the sum of the values of individual cells P_{MPP} and the total power extracted from the string, illustrated in equation 2.5.

$$P_{\text{loss,mismatch}} = \sum_{i=1}^N P_{MPP,i} - P_{\text{array}} \quad (2.5)$$

Partial shading creates differences in the irradiance received by the PV cells[12]. Each section of the array may receive different irradiance, creating mismatch losses. Partial shading is caused by physical shadows that block irradiance from reaching sections of the array. This is the primary cause of mismatch losses in series connected strings.

Partial shading effects and non-uniform irradiance effects can be mitigated through the use of bypass diodes[12]. These diodes are placed in parallel across groups of cells in reverse-biased configuration. If this group of cells becomes limited by shading or non-uniform irradiance effects, the current will be maintained, but due to the IV curve characteristics, the voltage across this group will drop. When the voltage drops below the forward-drop threshold of the bypass diode, the diode becomes forward-biased and starts conducting. This removes the under-performing cells from the current path and prevents damage to those cells.

When a bypass diode activates, the PV system loses the power contribution of those under-performing cells, the output voltage will drop and power is reduced, but damage to those cells from hot-spot heating is negated. This creates a discontinuity in the IV curve and results in multiple local maxima occurring in the PV curve[12].

Non-uniform irradiance effects on PV cell IV curves is generally not an issue in conventional PV systems such as solar farms or roof-tops. PV systems are designed to minimise the effects of partial shading, and PV arrays are arranged on a single plane. These effects become more pronounced in automotive systems, with a curved mounting surface and frequent partial shading events[1]. MPPT architectures fail to manage these issues and alternative architectures are required to maintain system integrity.

2.3 MPPT Algorithms

An MPPT is required for PV systems to maintain the operating point of a PV array at V_{MPP} and I_{MPP} [5]. This dynamic operating point requires constant supervision to ensure that the PV system is extracting maximum power.

An MPPT system consists of three key elements. A DC/DC converter adjusts the operating point of the PV array. A control unit reads sensor data, runs the MPPT control algorithm, and generates the required PWM inputs for the DC/DC converter. Sensing circuitry measures PV voltage and current to provide control feedback[5].

Two core algorithms are perturb and observe (P&O) and incremental conductance. P&O adjusts the operating voltage of the PV system and observes a change in power[5]. Incremental conductance searches for the point in which $\frac{dP}{dV} = 0 \rightarrow \frac{dI}{dV} = -\frac{I}{V}$, this is where P_{MPP} occurs[5]. These are simple algorithms that are very effective in finding the ideal operating point under ideal conditions[5]. However, these algorithms assume a single peak in the PV curve, struggle with partial shading, and are likely to converge to a local MPP[12].

These algorithms assume that the PV curve has a single maximum. Partial shading introduces multiple PV peaks. P&O and incremental conductance may converge to a local peak rather than reaching global maxima[12]. This behaviour can be partially rectified by periodically sweeping the entire voltage range of the PV array and calculating a PV curve to identify peak locations[12]. This has several issues. Whilst this sweep occurs, power is not being optimally generated, therefore the sweep can only be ran intermittently to mitigate static partial shading maxima. In frequently changing conditions, a voltage sweep is likely not help much, due to maxima occurring for short periods of time.

MPPT algorithms are critical to extract maximum performance from a PV system. More complex algorithms exist that help mitigate local maxima traps and perform faster navigation to the optimal operating point. These algorithms are based on a concept similar to the two presented here. The implementation of low-noise sensing and minimal converter ripple is essential for accurate performance for all MPPT algorithms.

2.4 Power Converter Topologies

To control the operating point of a PV system connected to a DC load such as a battery, a DC/DC converter is used[5]. By adjusting the duty cycle, the converter controls the input impedance seen by the PV array, allowing the array to be loaded at P_{MPP} [4].

2.4.1 Non-isolated Converters

Non-isolated converters can be split into three families. Buck converters are used to lower the output voltage below the input voltage[4]. Boost converters raise the output voltage above the input voltage[4]. Buck-boost converters are capable of creating a voltage that is higher than or lower than the input voltage[4].

Topology Selection

Choosing a topology depends on the voltage of the PV array and the desired output voltage. If V_{PV} is strictly less than V_{bus} , then a boost topology is ideal[4]. Likewise, if V_{PV} is strictly greater than V_{bus} , then a buck topology is ideal[4]. If the input and output voltages overlap, then a buck-boost topology may be necessary to ensure proper operation under all conditions[4]. It is especially important to consider how the voltage of the PV system will change during use and, if connected to a battery, the wide range of the output voltage.

Boost Converters

For reasons discussed in chapter 3, a non-isolated boost converter topology will be used for the LV MPPT design in this thesis. Therefore, it is important to understand how asynchronous and synchronous boost converters work. The principle of operation is similar for most non-isolated converters, sharing comparable design principles and power loss mechanisms. Figure 2.2 is a basic asynchronous boost converter.

Principle of Operation

Boost converters are able to increase the output voltage above the input voltage by storing energy in the inductor[4]. The ratio of $\frac{V_{out}}{V_{in}}$ is determined by the duty cycle of the PWM signal applied to the gate of the switching element. This relationship is given by equation 2.6. When the switch is on, current is drawn through the inductor into the ground, storing energy

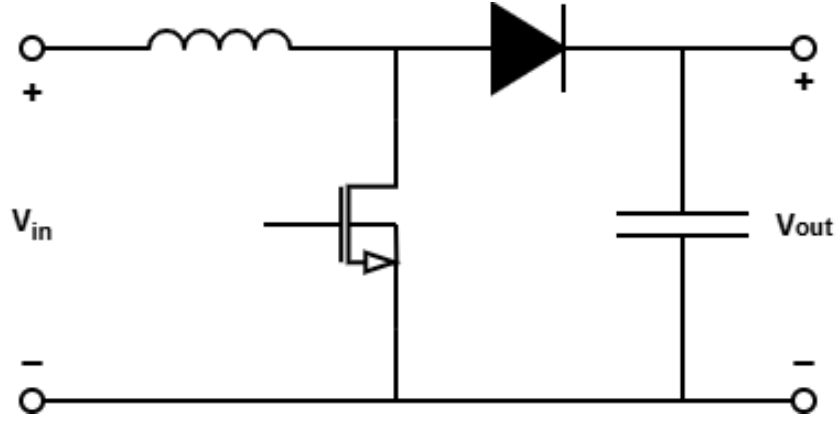


Figure 2.2: Asynchronous boost converter

in the magnetic field of the inductor. The diode prevents the output from feeding back into the ground when the switch is on. When the switch turns off, the inductor uses the energy stored in its magnetic field to maintain the current through it by inducing a voltage across its terminals. This voltage is added to the input source voltage, creating a larger voltage on the output. The capacitor stores energy and releases it whilst the switch is on, to prevent an output ripple. The magnitude of the induced voltage across the inductor is determined by how long the inductor is charged for; therefore, controlling the switch duty cycle allows for accurate adjustment of the output voltage.

$$V_{out} = \frac{V_{in}}{1-D} [4] \quad (2.6)$$

Asynchronous and Synchronous Converters

Asynchronous converters can be distinguished from synchronous converters by the existence of a diode to prevent current from flowing from the wrong source. This diode is the main source of inefficiency in asynchronous converters because they have a forward-voltage drop. The power loss due to this drop is given by $P_{diode} = I_{out} \cdot V_F$. To mitigate the need for a diode, another switching element is placed instead of the diode. The switching elements are driven with a complementary pair of PWM signals, preventing back-flow when the primary switching element is on. Dead-time is injected between the active period of the signals to prevent a direct short between the output and the negative reference. This is illustrated in figure 2.3.

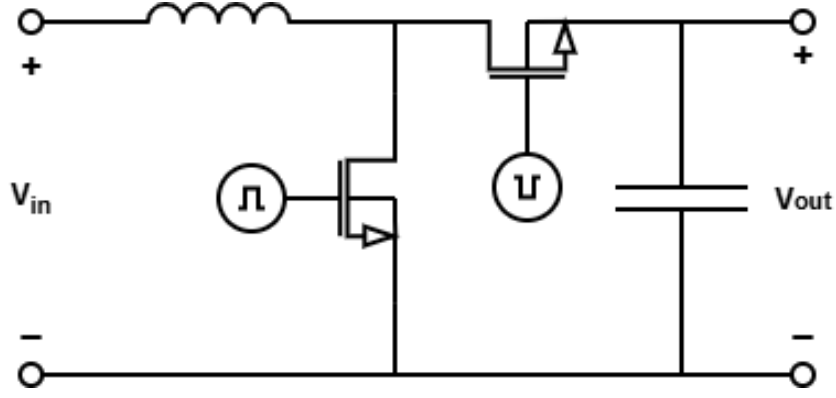


Figure 2.3: Synchronous boost converter

Ripple Current

Although the principle of boost converter operation is relatively simple, designing an efficient boost converter involves careful consideration of a variety of features.

The ripple current (ΔI) is the difference from peak to peak of the current through the inductor. It is caused by the switching action of the converter that alternates the inductor between storing and releasing energy. The ripple current can be estimated using equation 2.7. Increasing inductance and converter switching frequency reduce ripple current.

$$\Delta I \approx \frac{V_{in} \cdot D}{L \cdot f_{sw}} [4] \quad (2.7)$$

A large ripple current introduces higher conducted losses in the DC/DC. The conducted losses are due to $P = I_{RMS}^2 R$, and although the average current at any given ΔI may be the same, I_{RMS} will be higher at larger ΔI . A smaller ΔI also allows smaller input currents to be realised before the inductor enters the discontinuous conduction mode (DCM). DCM occurs when the inductor completely discharges before the next cycle begins, causing non-linearity, larger current ripple, and increasing inductor losses. Additionally, current ripple can pass through the DC/DC converter and contribute to higher $\frac{di}{dt}$ seen at the output capacitor, leading to component stress and contributing to output voltage ripple. Current ripple can also be passed through the MPPT current sensor leading to inaccurate measurements and noise.

Typically, a design target of $\Delta I \approx 10\%$ is chosen. Increasing the inductance and switching frequency reduces ripple current, but can lead to larger conducted losses from the DC resistance of the inductor, and switching losses in the switching elements.

Ripple Voltage

The voltage ripple (ΔV) is the variation in output voltage around its average value. Similarly to the current ripple, it is also caused by the switching action of the converter due to the fact that the output capacitor must supply energy whilst the inductor is charging. Ripple voltage can be calculated using equation 2.8.

$$\Delta V_{out} \approx \frac{I_{out} \cdot D}{C \cdot f_{sw}} [4] \quad (2.8)$$

A larger ripple will cause an increase in component stress, especially in the output capacitor. Voltage ripple is an important consideration considering that the boost converter for this thesis will be directly connected to the internal voltage bus. Introducing a large AC component to a system of sensitive embedded electronics is undesirable.

A reasonable design target for ΔV is 1 – 3% of the output voltage. Ripple voltage is dependent on capacitance and switching frequency. A small ripple voltage is achievable by increasing the output capacitance. Unlike increasing inductance, increasing capacitance does not introduce more loss to the system, and the addition of multiple capacitors can reduce the ESR leading to lower conducted losses.

Switching Frequency

An important part of DC/DC design is choosing the switching frequency. As seen in the discussion of current and voltage ripple, increasing the switching frequency decreases current and voltage ripple, resulting in better performance for the overall design. Additionally, as seen in equation 2.7 and 2.8, increasing the switching frequency also allows for a reduction in the size of the components whilst maintaining the same performance. This is especially important in space constrained systems for maximising power density. However, switching losses directly depend on the switching frequency. Switching losses occur when the switching element transitions between states. At each transition, some energy is dissipated in the switch as either the voltage or the current in the switching element is simultaneously high for a brief period. This is shown in equation 2.9.

$$P_{sw} \approx \frac{1}{2} VI(t_r + t_f) f_{sw} [4] \quad (2.9)$$

Increases in switching frequency also increase gate-charge losses. For a MOSFET switching element, in every switching cycle, a small amount of energy is lost charging the gate to activate

the switch, this is modelled in equation 2.10.

$$P_{gate} = Q_g V_g f_{sw} [4] \quad (2.10)$$

FET Selection

Choosing the right FET technology can help mitigate switching losses. Silicon MOSFETS are widely used in power electronics. They are a very mature technology that is fairly low-cost and gate-driving is simple. This makes them ideal for mass-manufactured products. Although silicon FETS have low $R_{ds,on}$ indicating low conducted loss, they have relatively slow rise and fall times, as well as a high gate charge, meaning that the switching frequency is limited to about 20 kHz to 200 kHz[18]. Silicon carbide MOSFETS (SiC) are more costly but can withstand higher voltages and have far better switching performance[18]. Gallium nitride (GaN) devices have very low gate charge and output capacitance with extremely fast switching up to 1 MHz, but are relatively expensive and require sensitive layout considerations[18]. The choice of FET is extremely application dependent, and there is much more to consider than switching losses when choosing an FET. Boost component selection reasoning is expanded on in the design implementation in chapter 4.

2.4.2 Isolated Converters

In many systems requiring DC/DC converters, isolation is a strict requirement[4]. This is true for the Sunswift SR8 HV EV battery, where galvanic isolation to the HV system is required for safety and to maintain a galvanically disconnected HV system when not in use.

There are a number of topologies that can be used for isolated DC/DC conversion. Some examples include flyback converters, forward converters, half-bridge converters, and full-bridge converters[9]. Like non-isolated converters, the choice is application dependent. This section will briefly examine these converters, as an isolated DC/DC converter is required for this thesis design to enable HV battery charging from the PV system. It is important to note that the isolated DC/DC converter will be connected to the PV system through a LV DC-link (the LV bus).

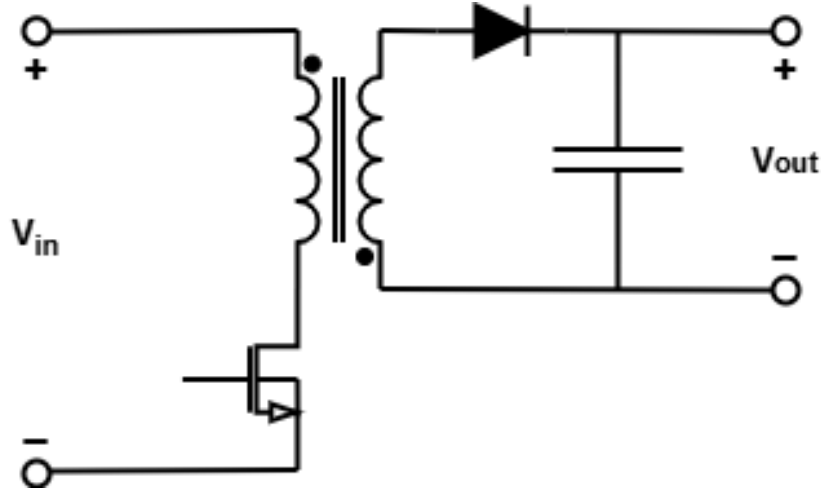


Figure 2.4: Flyback Converter

Flyback Converters

Flyback converters use a coupled inductor, as shown in figure 2.4. The input and output voltage relationship is similar to an inverting buck-boost converter with the addition of a turn ratio, displayed in equation 2.11.

$$V_{out} = \frac{N_s}{N_p} \frac{D}{1-D} V_{in} \quad (2.11)$$

The operation is similar to that of the boost converter, where whilst the switch is on, current flows through the inductor on the primary side, storing energy in the magnetic field (specifically in the air-gap of the magnetic core)[4]. During this period, the diode on the secondary side is reverse-biased, a voltage is induced in the secondary winding, but no current flows in the output. When the switch is turned off, the primary and secondary winding voltages reverse polarity, allowing the secondary diode to become forward-biased so current can flow through the output. Like the boost converter, the capacitor supplies energy during the switch 'on' period to ensure a stable output.

Flyback converters are very simple to operate and control, and it is possible to achieve very high output voltages with a high turn ratio. Since the energy is stored in the air-gap of the magnetic core, the practical power throughput of this topology is limited before major efficiency losses and ripples are realised[4].

Forward Converters

Forward converters are similar to flyback converters, but transfer energy whilst the switch is on and off[9]. Forward converters use a transformer with a single magnetic core, but the primary side generally contains two sets of windings. When the switch is turned on, energy flows through the primary winding, inducing a change in flux in the core and, therefore, a voltage across the secondary winding. The secondary side contains two diodes, one of which will be forward-biased, allowing current to flow in the output. The same process occurs when the switch is turned off, except that the secondary winding voltage is inverted, so that the other diode becomes forward-biased. This operation causes flux to accumulate in the core, leading to magnetic saturation and zero flux linkage. This is why the primary side has another set of windings, which apply a reverse voltage to the core during the switch off period, forcing flux back to zero. This is common, but there are other methods that are used to reset the magnetic flux in the core.

Forward converters require a slightly more complex design to prevent magnetic core saturation, but can provide a stable output voltage and maintain efficiency at higher power[9].

Bridge Converters

Bridge converters perform DC/DC conversion by creating an AC voltage across the mutually coupled transformer windings and then rectifying the AC voltage induced on the secondary winding[4]. These can be separated into full-bridge and half-bridge converters. The name indicates the method of creating an AC voltage. A full-bridge (also known as an H-bridge) uses 4 switches in two totem-poles, with the primary winding connected between the middle of each pole. The switches are then alternated in two configurations to rapidly switch the voltage polarity on the primary winding, resulting in continuous energy transfer through the transformer. A half-bridge transformer operates with two switches but can only provide the primary winding with half of the input voltage. Since more current is needed at a lower voltage for the same power as a full-bridge converter, half-bridge converters suffer higher conducted losses. A pair of diodes or synchronous switches is used to rectify the AC voltage, and an LC filter provides smooth DC output[4].

Although bridge converters are more complex than other converters previously discussed, they

offer superior performance characteristics because energy is continuously transferred through the transformer, rather than being stored and released. This results in higher efficiency, lower ripple, lower conducted losses, and higher practical power through-put[4].

Summary

This section has discussed a few non-isolated topologies that could be implemented for the SR8 HV link, and provided practical advantages and disadvantages for each. Although this section of design is left for thesis B, it will likely involve implementing a full-bridge or half-bridge converter topology with a synchronous rectifier to maintain maximal efficiency, given the 750W peak PV power.

2.5 Existing Solutions

There are two dominant MPPT architectures, SMPPT systems with interconnected arrays and distributed MPPT systems. This section evaluates the advantages and disadvantages of each solution architecture.

2.5.1 SMPPT Systems

A single MPPT (SMPPT) system involves having the arrays in a PV system connected in series and/or parallel and connecting these to an MPPT.

This is the most simple and low-cost solution for integrating an MPPT into a PV system. Integrating an SMPPT system into a DC or AC system is much simpler than other solutions, as there is only one entry point to consider. It is also highly efficient, more so than many other solutions, given that the PV system operates under uniform conditions.

The primary drawback of a SMPPT system is the fact that it operates very poorly under non-uniform conditions. This means that the PV system designer must carefully plan their PV installation to limit partial shading, difference in panel angles, and other non-uniform factors to prevent major mismatch losses.

Fundamentally, a SMPPT solution means that the system can use only a single operating point

for all PV cells[1]. As soon as one or more PV cells in a series string begin to operate in a smaller I_{MPP} , the entire string becomes limited by that cell, creating large mismatch losses[7]. Similarly, parallel-connected strings will only be able to operate at a single terminal voltage. If one string has optimal power output at a different voltage, then that will not be satisfied by the controller and result in efficiency decreases for that string.

SMPPT systems are ideal for PV installations that generally operate under uniform conditions. They require more planning and suffer major losses in dynamic conditions, but overall are simpler to implement.

2.5.2 Distributed MPPT Systems

Distributed MPPT systems use multiple MPPT modules to segment the PV system and operate each section with a different operating point. This is in direct contrast with SMPPT systems, only offering the ability to operate at a single point.

String and Module Level MPPT's

String and module level MPPT systems refer to having multiple MPPT's for a single PV system, rather than interconnecting all the arrays into a SMPPT. The PV system designer may choose to have one MPPT per module (an array with multiple strings), one MPPT per series string, or even choosing to segment strings into smaller strings with individual MPPT's[1].

As you increase the number of MPPT's in the system, the cost will rise significantly. There may also be increased integration risks if multiple MPPT's are charging the same battery, depending on the specific MPPT products or the MPPT design. There is also increased mechanical integration complexity due to a larger number of devices present. However, every additional MPPT in the system adds additional robustness to dynamic conditions by adding operating point flexibility for the section it controls. It also allows the system to operate with sections at different angles and to minimise partial shading effects on the global P_{MPP} [13].

DC Optimisers

Some literature refers to string and module level MPPT's as DC optimisers. This thesis recognises DC optimisers as distributed MPPT method applied on sections of cells inside a series string.

DC optimisers are small unidirectional or bidirectional DC/DC converters that are connected to groups of cells inside a series string, illustrated in figure 2.5. These devices can be used in conjunction with a primary string or module MPPT, or without an external MPPT depending on the configuration. The devices in figure 2.5 are intended to be used with an external MPPT. These DC optimisers either inject or extract power from cell groups[10]. The injection case is intended to support the weak cells in the PV string by injecting current into blocks of under performing cells, to allow the cell to still contribute some power, and to prevent the string from becoming current limited. These are used instead of diodes to circumvent the undesired loss in a diode and to allow the cell to still contribute some power to the system. The extraction mode is similar and allows the PV string to operate at the current of the weakest cell, and the DC optimiser extracts the difference in power to prevent mismatch losses. In some configurations, the final DC optimiser may act as a string based MPPT to mitigate the need for an external string based MPPT.

DC optimisers are widely regarded as an effective solution to solving mismatch losses in PV systems. However, they offer considerable integration complexity, especially for existing PV arrays[1]. DC optimisers require access to intermediate connection points in the PV string and require arrays without bypass diodes. These PV arrays are commercially available, but the choice of array will be more limited. The other major drawback is electrical and mechanical complexity. Integrating multiple DC optimisers per series string means figuring out mounting locations on the array or nearby, and designing the external harness. In space limited systems, this is likely not an option.

2.5.3 Summary

SMPPT's, string/module based MPPT's, and DC optimisers are key MPPT system architectures that cover a wide range of use cases. SMPPT systems are well-suited to static installations that

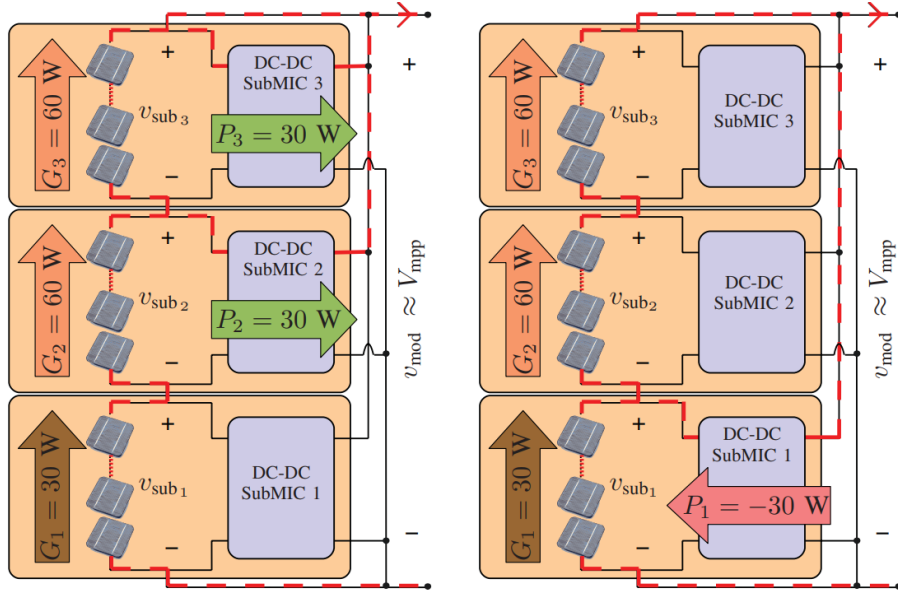


Figure 2.5: DC optimiser configuration[10]

have had extensive planning to mitigate partial shading effects. Distributed MPPT's are ideal for PV installations with static array differences and for dynamic environmental conditions. Distributed MPPT systems are very customisable, with the ability to control the operating point of a single cell, string, or module to improve energy extraction under non-uniform conditions. They have increased integration complexity, but are far more robust. They are also far more scalable and reliable due to having multiple control modules[6].

The SR8 PV system will operate under highly dynamic operating conditions, and the PV arrays are not on a single plane. This analysis indicates that the optimal approach for the MPPT architecture is a distributed MPPT architecture. Chapter 3 explores the SR8 system in detail and proposes a solution to optimise PV system efficiency whilst maintaining an acceptable level of integration complexity.

Chapter 3

Proposed Solution

3.1 System Requirements and Constraints

This thesis is focused on developing a MPPT system for the PV arrays on SR8. It is imperative to gain a thorough understanding of the existing electrical architecture of SR8 to design a system that integrates safely and perform efficiently. The aspects pertaining to the design of an MPPT system include battery specifications, PV array specifications, and preferred telemetry format.

Like most EV's, SR8 utilises a dual battery architecture. A HV 45 kWh , 648 V_{nom} battery delivers power to the combined 332 kW motors. A LV 2.3 kWh , 46.8 V_{nom} battery powers embedded components inside the car, and can remain on whilst the HV battery remains off. The MPPT must be able to charge both the LV battery and the HV battery. The specifications of these batteries are described below in table 3.1.

Specification	LV Battery	HV Battery
Cell	INR-21700-P45B	INR-21700-P50B
Configuration	13S10P	180S14P
Capacity	2.34 kWh	45.36 kWh
V_{min}	32.5 V	450 V
V_{nom}	46.8 V	648 V
V_{max}	32.5 V	756 V

Table 3.1: SR8 battery specifications

SR8 has 5 PV arrays positioned on the exterior body panels. These PV arrays are each a single series string of cells. These are comprised of SunDrive G12-210 SD-HJT cells. These are heterojunction solar cells and have the highest efficiency for mass-produced solar cells. For the purposes of this thesis, it is enough to recognise that these have a conventional IV curve. The key per-cell characteristics are outlined below in table 3.2.

Characteristic	Value
Efficiency	$25.1\% \pm 0.1\%$
P_{MPP}	5.53 W
V_{MPP}	0.673 V
I_{MPP}	8.22 A
V_{OC}	0.744 V
I_{SC}	8.66 A
Temperature Coefficient of P_{max}	$-0.258\text{ \%}/^{\circ}\text{C}$
Temperature Coefficient of V_{OC}	$-0.27\text{ \%}/^{\circ}\text{C}$
Temperature Coefficient of I_{SC}	$0.055\text{ \%}/^{\circ}\text{C}$

Table 3.2: Sundrive G12-210 SD-HJT cell characteristics [15]

The PV arrays are positioned on the front (F), roof (R), and back (B) of the vehicle. The positioning of these arrays is illustrated in appendix 1. There is one large PV array on the front, with two on the rear panel and 2 on the roof panel. It is important to mention that the arrays have been segmented such that the curvature of the arrays is minimised within series string. The effective angle between each array can be as large as 20° . The electrical parameters of each array and G12-210 SD-HJT cell counts are listed in table 3.3. PV cells are arranged in series for each array.

The maximum power generated by the SR8 PV system is 769 W , however, this number will be lower because I_{MPP} being limited due to the fact that the arrays are subject to a glass lamination process to be protected from physical damage whilst in the car. This maximum power will also not be achieved due to the effective irradiance difference between arrays and losses in the MPPT and harness.

Array	Cells	V_{OC} [V]	I_{SC} [A]	V_{MPP} [V]	I_{MPP} [A]	P_{MPP} [W]
F1	41	30.50	8.66	27.59	8.22	226.81
BL	24	17.86	8.66	16.15	8.22	132.77
BR	24	17.86	8.66	16.15	8.22	132.77
R1	32	23.81	8.66	21.54	8.22	177.03
R2	18	13.39	8.66	12.11	8.22	99.58

Table 3.3: SR8 PV array characteristics

The SR8 embedded system communicates using FD-CAN, a modern version of CAN designed for higher bandwidth and larger payloads. CAN is a bus-based differential communication protocol which is commonly used in automotive systems. This MPPT design must communicate with other systems in the car and transmit telemetry over FD-CAN.

The Sunswift Racing solar team has contributed a soft requirement, in the form of being able to experiment and test their own MPPT algorithms, and test there solar panels off-car. Therefore, this design should include the ability in firmware to connect to a simple web-based interface to deploy experimental algorithms and receive telemetry from each MPPT channel.

There is also strict integration requirements that dictate the maximum size and mounting location of the MPPT in SR8. The MPPT design including the HV component shall be no more then 30cm by 20 cm by 10cm. It will also be mounted in a section of the car exposed to water ingress, so an appropriate IP target must be decided upon and satisfied by this design. Integration of electronic components in automotive environments requires vibration mitigation, this was recognised as the primary fault mechanism in the 2025 BWSC race for SR7. Therefore anti-vibration measures should be taken when mounting the MPPT and for securing internal components.

3.2 Solution Architecture

Gaining an understanding of the system requirements allows an educated decision to be made for the MPPT architecture. The concept of mismatch losses was introduced in chapter 2. To

mitigate mismatch losses caused by differences in effective irradiance angle and due to partial shading of panels, a distributed MPPT system will be implemented on a single board. For each PV array (string) on the car, there will be a separate DC/DC converter performing it's own MPPT algorithm. This string based approach was chosen to balance integration complexity with mismatch performance. An alternate solution would be connecting PV arrays in series or parallel, and using an SMPPT. This solution is simpler but yields much higher losses due to the difference in effective irradiance between panels and is more susceptible to partial shading effects.

This distributed MPPT solution will feed power into the LV bus. A micro-controller will handle software-based protection measures, MPPT control, and output charge state control. There will also be a HV DC/DC to facilitate HV battery charging when the LV battery is at capacity. This architecture is illustrated in figure 3.1.

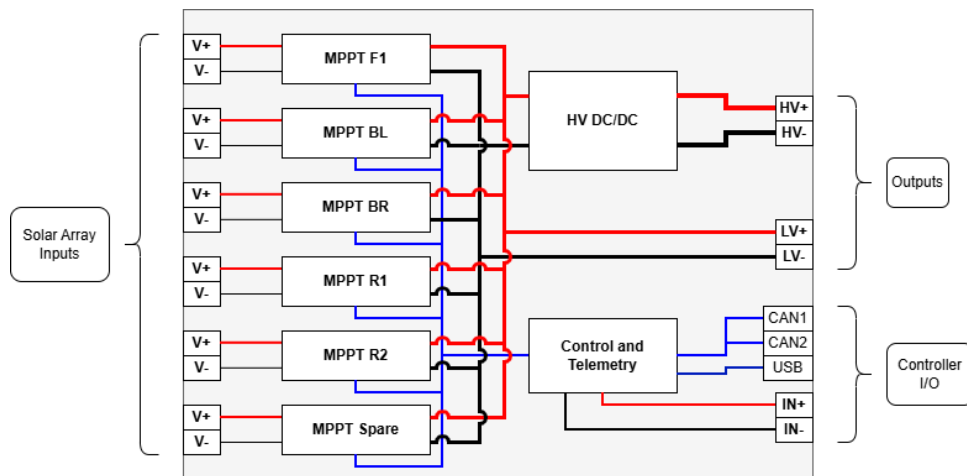


Figure 3.1: Block diagram of solution architecture

3.3 Power Stage Topology

In table 3.1 and table 3.3 it can be noted that the minimum LV bus voltage is 32.5 V, the maximum PV array voltage is 30.50 V, and the maximum PV array MPP voltage is 27.59 V. This indicates that the output MPPT voltage is strictly above the maximum MPPT input voltage. As temperature increases past STC, the input voltage lowers, meaning that this statement is still valid. It is unlikely for the panels to decrease below STC temperature conditions during operation.

Due to the relatively low-voltage present in the LV MPPT system, the efficiency and complexity trade-offs for isolation are not feasible. Therefore, a GaN FET based synchronous boost converter will be used. This topology was chosen due to the input and output voltage ranges, as well as holding several efficiency advantages over asynchronous boost converters. GaN FET technology will be used to increase the converter switching frequency whilst limiting switching losses due to the low gate charge characteristics. This allows for far smaller inductors and ripple current over other FET technologies such as silicon FET's.

3.4 Control Strategy Overview

MPPT control will be conducted independently for each MPPT channel. Initially, a simple hill-climbing control algorithm such as P&O will be used to target maximum efficiency in lab-based conditions. This will be paired with a low-frequency periodic current-sweep to combat local maxima traps during stationary solar charging. These algorithms will be executed on a single micro-controller with each converter receiving a complementary pair of synchronised PWM signals. The micro-controller will also be capable of injecting dead-time. To optimise converter efficiency, dynamic frequency changes may be incorporated.

The MPPT system will modularise algorithm's to allow for a web-based interface to select, modify, and deploy algorithm's at a user's request. This means that the MPPT control will be a two-layer system with a high-level module and optimisation and protection measures operating at a lower level.

Software over-current and over-voltage protection measures will be used for channel inputs and bus outputs. The MPPT will monitor battery bus voltage and regulate MPPT output current when the battery is approaching capacity. Once the LV battery reaches capacity, the MPPT will also transition to HV charging to maximise solar energy extraction. This will be done independently of BMS commands to prevent the battery from over-charging in the event that the BMS fails or communication fails.

3.5 Design Advantages

This design approach was carefully chosen to mitigate the disadvantages associated with existing solutions and to integrate easily with the existing SR8 architecture.

The multi-channel architecture expresses considerable benefits when compared to a MPPT configuration with PV arrays in series or parallel. It allows for the effective irradiance mismatch between arrays to not bottle-neck power output. This is because MPPT algorithms will navigate to the most efficient operating point for each PV array, not the combined operating point. For the same reason, it also mitigates efficiency losses due to partial shading. Another advantage is that it allows for precise telemetry for each array, facilitating array fault-finding and testing.

Existing MPPT solutions contain a single output, often to a commonly used battery voltage. This design incorporates dual outputs. The primary outputs will be used for charging the LV battery. However, 3 hours of full sun is enough to almost reach LV capacity, therefore a HV charging system will be incorporated to utilise the PV array when this is the case. This also means that an external LV/HV DC/DC link will not be required.

An internal MPPT design offers several integration advantages, and allows full control of telemetry and tight integration with existing systems.

Chapter 4

Preliminary Work

This section will detail the design work that has been undertaken in thesis A. This design work follows the proposed solution in chapter 3.

4.1 MPPT Power Stage Design

A core focus of the MPPT design is the LV power stage. This is one of the major factors that will limit the PV system efficiency. The proposed solution dictated the chosen topology (Synchronous boost converter) and the reasons why it was chosen. This section will provide reasoning behind why specific design choices were made. In chapter 2 a detailed analysis into synchronous boost converters was conducted, revealing key design aspects that this design takes into consideration.

4.1.1 Boost Converter Design Parameters

Ripple Current and Voltage Targets

The first important parameter to choose is ripple voltage, ΔV . To maintain output stability and ensure sensitive electronics on the LV bus are not effected, a value of $\Delta V = 2\% \cdot V_{out,nom} = 0.936 \text{ V}$ was targeted. Likewise, ripple current was chosen to be relatively small, at $\Delta I = 10\% \cdot I_{in,max} = 0.822 \text{ A}$. Inductance, capacitance, and switching frequency will be chosen to keep these values under these thresholds.

Switching Frequency

Following the ripple threshold selection, a nominal value for f_{sw} should be selected. A value of 500 kHz will be picked for inductance and capacitance calculations. This value yielded acceptable efficiency in preliminary testing, which is discussed in a later section. The switching frequency is likely to be a dynamic variable for efficient operation at light loads.

Inductance Calculation

To keep the LV MPPT design simple, a single MPPT will be designed, and used for all MPPT channels. Choosing an inductor with multiple values for a single footprint is desired in case these values can be reduced. Inductance is determined using equation 4.1. Nominal battery voltages and the MPP conditions of the longest PV string represent an acceptable point in which to calculate the inductance. With $V_{in} = 27.59$ V, $V_{out} = 46.8$ V, and $D = 0.41$. This yields an inductance of 27.59 μH .

$$L = \frac{V_{in} \cdot D}{\Delta I \cdot f_{sw}} \quad (4.1)$$

Capacitance Calculation

The required capacitance can be calculated using equation 4.2. Increasing capacitance beyond the required capacitance has negligible downsides, apart from an increase in size. Therefore, calculated capacitance will be taken as a rough baseline value when choosing components. With $P_{out} = 226.81$ W, $V_{out} = 46.8$ V, $I_{out} = 4.85$ A, and $D = 0.41$, the minimum output capacitance is only 4.25 μH .

$$C = \frac{I_{out} \cdot D}{\Delta V \cdot f_{sw}} \quad (4.2)$$

4.1.2 Chosen Components

Following minimum value calculations, components can be chosen that match or exceed those thresholds.

Inductor Selection

Inductors have multiple variables to consider when making a selection. The most obvious is inductance. This value will decrease as current increases and builds a magnetic field in the core. The saturation current indicates this decrease and is typically the value at which the inductance

has dropped to 80% of the original value. Since our design will draw high current during normal operation, I_{sat} should remain above 8.22 A. The core material, size, thermal performance, DC resistance, and EMI shielding are all factors to consider.

For this design, the Coilcraft SER2211[2] series was identified as a suitable range of inductors to select from. These are flat-wire inductors in a relatively small (22.5 mm) package. They are magnetically shielded, with a ferrite core for stable inductance over a wide current range. This series contains 7 inductance values from $3.0\mu H$ to $47\mu H$ which all use an identical footprint, allowing inductance changes after the board has been ordered.

Specifically, the SER2211-333MED $33\mu H$ inductor will be used initially for all channels, with the $27.3\mu H$ and $12.3\mu H$ versions being purchased as spares for testing. The SER2211-333MED features a 20% inductance drop at 8.0 A. This is likely the maximum current the inductor will experience after accounting for the glass lamination losses in irradiance. Therefore, the inductance at this current is roughly $26.4\mu H$. This is slightly less than the target inductance, but still yields an acceptable input current ripple of about 0.86 A. This ripple is large enough to cause inaccuracies in the current measurement, so appropriate filters will be considered. This inductor also has a low DC resistance of $8.95\ m\Omega$.

Capacitor Selection

The capacitor selection was straightforward as a result of the small minimum capacitance required to reach the target threshold. A total capacitance $48.84\mu F$ was chosen in the form of 2 x $22\mu F$ capacitors, 2 x $2.2\mu F$ capacitors, and 2 x $220nF$ capacitors. The $22\mu F$ capacitor is a SMT aluminum electrolytic capacitor (EEEFK2A220P[11]) with a 100 V rating. The $2.2\mu F$ capacitors and $220nF$ capacitors are X7R ceramic capacitors rated to 100 V.

This combination of capacitors was chosen for the ability to provide bulk energy storage to limit ripple current, and to provide a low ESR path for high frequency voltage transients through the ceramic capacitors.

FET Selection

Since the switching frequency of this design has been set fairly high to allow the inductor to be small, an appropriate FET selection must be made. Differences between various FET technology is explore in chapter 2, and the decision has been made to proceed with GaN FET's.

Selection of the GaN FET was based on a combination of gate drive complexity, gate charge, $R_{ds,on}$, propagation delay, and drain/source voltage ratings. The Texas Instruments LMG GaN FET series contains a variety of GaN FET packages. LMG2100R026[16] was chosen as the GaN FET for this boost design due to the package including an integrated gate driver and dual FET's in a totem-pole configuration, greatly simplfying the design and reducing risk of layout error. This device is capable of switching up to 10 MHz, has schmitt triggered logic-level compatible PWM inputs, 53 A drain current rating, 93 V voltage rating, 2.6 mΩ $R_{ds,on}$, a gate charge of 12nC, and rise and fall times of about 35ns.

4.1.3 Loss Analysis

There are several loss mechanisms. This section will analyse the magnitude of primary loss sources such as conducted loss and switching losses, for F1 array conditions.

Conducted Loss

Conduction loss comes from all parts of the boost converter. We can estimate the conducted loss by calculating the average DC resistance between V_{in} and V_{out} . The resistance of traces is not considered. Losses are summarised in equation 4.3. Assuming an ESR of 1mΩ, $P_{conducted} = 0.78\text{ W}$ at an input current of 8.22A.

$$P_{conducted} = I_{in}^2 \cdot R_L + I_{in}^2 \cdot R_{ds,on} + \Delta I_{rms}^2 \cdot ESR[4] \quad (4.3)$$

Switching Losses

Switching losses have been previously explained in chapter 2. Using equation 2.9 to calculate the switching loss for the low-side FET, where $V = V_{out} = 46.8\text{ V}$, $I = I_L = 8.22\text{ A}$, $t_r = t_f = 35\text{ ns}$, and $f_{sw} = 500\text{ kHz}$, it is found that $P_{sw} = 6.7\text{ W}$. This is large loss, at 2.9% of the total power. Switching frequency may be adjusted to reduce this loss during testing.

Gate charge loss will not be calculated since the gate driver is packaged with the FET devices. However, the LMG2100R026 datasheet provides the total operating current of VCC at 500 kHz. Therefore $P_{driver} = V_{CC} \cdot I_{CCO} = 5 \cdot 15m W = 0.75W$.

Loss Summary

The combined losses are given by $P_{tot} = P_{conducted} + P_{sw} + P_{driver}$. There are other forms of loss in the system, but these calculations grant a reasonable expectation for actual efficiency. The total power loss is $P_{tot} = 8.23 W$. At an input power of $P_{in} = 226.81W$, the output power is expected to be $P_{out} = 218.58$ with an efficiency of 96.37%. This is a good starting point. This efficiency will be greater at lighter loads, and optimisations, such as dynamic frequency, will greatly improve the overall efficiency across the load range.

4.2 Auxiliary Circuits

This section will outline preliminary design and selection for a variety of systems across the design that are necessary to operate the LV MPPT system.

4.2.1 Micro-controller

A STM32H743ZIT6[14] micro-controller was chosen to implement control for the 6 MPPT channels. This was chosen primarily due to the available timer hardware. HRTIM1 is a high-resolution 16 bit timer with 10 outputs, or 5 complementary outputs. 16 bit 480 MHz resolution at 500 kHz gives a minimum time-step of 2.08ns and duty cycle resolution of 0.1%, allowing the MPPT to have fine control over the duty cycle, as well as being able to inject dead time using software. Investigation and preliminary firmware is required to ensure that the device has enough program memory to perform 6 MPPT algorithms simultaneously, external RAM and flash memory may be required.

4.2.2 CAN Transceiver

An isolated CAN transceiver was chosen, ISO1044BDR, and implemented according to the reference schematic. Details can be found in the appendix in the current schematic.

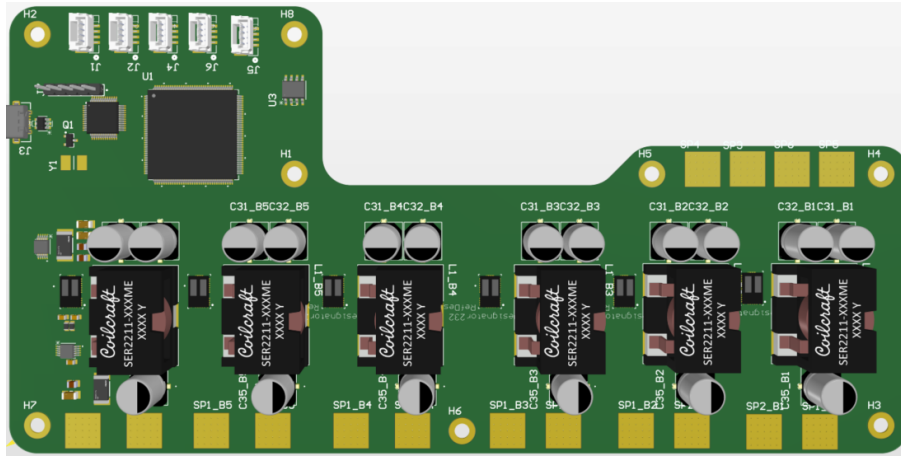


Figure 4.1: MPPT draft layout

4.2.3 Power Sensors

There are a total of 12 power sensors across the board, 2 for each channel. Given the number of sensors, an I2C power sensor (INA238AIDGSR) will be implemented to measure voltage and current. I2C has the added benefit of increased signal integrity, especially considering the number of switching elements on the board. They are currently implemented with $15m\Omega$ and $20m\Omega$ shunts. At an input current of 8.22 A and an output current of 4.84 A, these shunts dissipate 1.48 W. Lowering the shunt value to $5m\Omega$ means that the shunts will only dissipate 0.45 W, which is acceptable. Investigation must be done to find resistors of the right quality, and ensure the current sensor remains accurate enough for the MPPT algorithms.

4.3 Floor Planning

Estimating the total size of the design is helpful for planning on-car integration into SR8. Once the boost converter components and a majority of the auxiliary components were added to the schematic, a rough layout was placed with mounting holes to estimate size. This is shown in figure 4.1. The design is currently 200mm x 100mm x 13mm. This is excluding the enclosure, which will likely add 20mm in each axis to accommodate panel-mount connectors.

4.4 Design Validation and Results

Some basic validation has been conducted for the LMG2100R044. Dr. Ivan Perez-Wurfl and SPREE student The Vinh Nguyen have developed a LMG2100R044 development board that

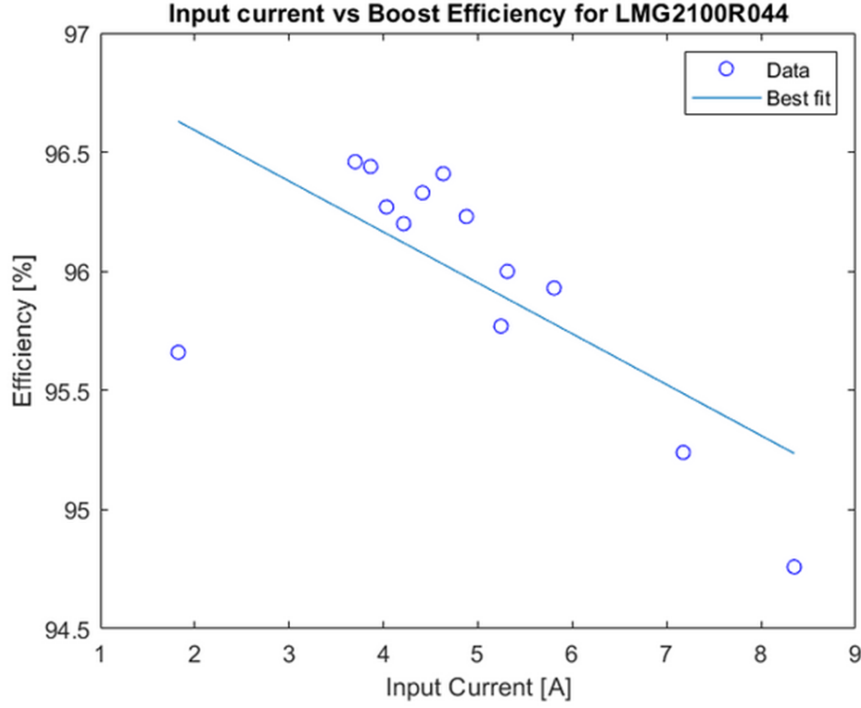


Figure 4.2: Input current vs boost efficiency

can be used in buck and boost configurations. They kindly shared this board with the Sunswift team for evaluation. The author was able to use this board alongside The Vinh Nguyen and Sunswift members to run some basic tests to determine the efficiency of the LMG2100R044 as a boost converter. This design had a lower inductance of $22\mu H$, and uses a different device from the same family, but the system is similar to the one that this thesis intends to implement. Figure 4.2 demonstrates the boost efficiency against the current. This is similar to the efficiency calculated in the loss analysis above. These results reinforced the design approach this thesis is using for the boost converters, and the efficiency can be improved on in thesis B.

Chapter 5

Project Planning

This thesis can be broken into 5 distinct stages.

1. System requirements and architecture.
2. LV MPPT design and test.
3. Firmware design and test.
4. HV design and test.
5. Performance evaluation and integration.

A detailed flow is displayed in figure 5.1 and these stages are listed with expected dates in the project timeline in figure 5.2.

The first stage is determining the system requirements that govern design decisions, and using these to develop an initial architecture. This is now complete and is discussed in depth in chapter 3.

The second stage is design of the LV MPPT modules. This is most of the way though now, with major design decisions being made, and the schematic and layout well underway. This has also been discussed in depth in chapter 4.

In thesis B, the focus will be testing the LV MPPT that should be completed over the term 1 break. In this term, firmware development and HV design of a full-bridge DC/DC converter will begin. Stages 3 and 4 should be completed around the start of term 3. This leaves the rest

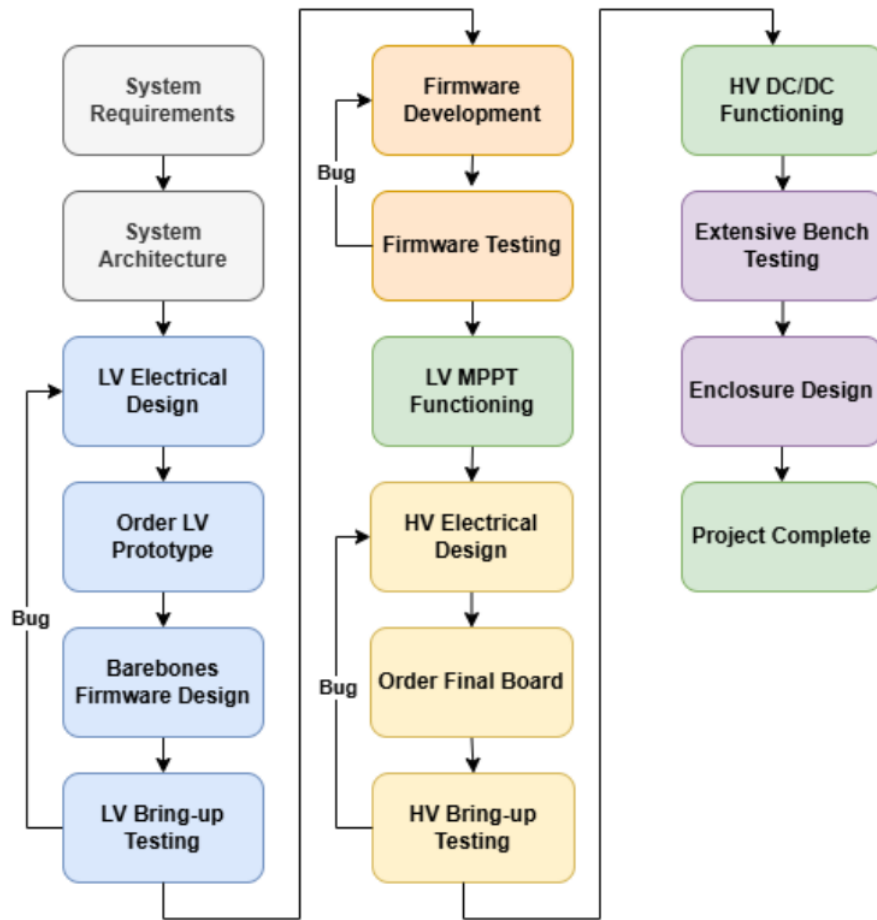


Figure 5.1: Project flowchart

of term 3 to integrating the design into the SR8 electrical and software network, and analysing the performance of the design for the final report.

Project Timeline	Term 1				Term 2			Term 3		
	February	March	April	May	June	July	August	September	October	November
System Requirements and Architecture										
LV Design and Test										
Firmware Design and Test										
HV Design and Test										
Performance Evaluation and Integration										
Documentation and Report Writing										

Figure 5.2: Project timeline

Chapter 6

Conclusion

In conclusion, the goal of this thesis is to develop an MPPT product to be used by the Sunswift team for SR8. This thesis explores and evaluates possible solutions for optimal PV system energy extraction under highly dynamic environmental conditions the system is operating under. This report has analysed how PV cells work, and the factors that impact the power they produce. A variety of converter topologies was also discussed, along with the fundamental problem of mismatch losses in dynamic PV systems. Existing solutions were analysed to find strengths and weaknesses that inform the proposed solution. The design philosophy behind the proposed solution and the reasoning for component and value choices in the preliminary work was also explored.

The use of a multi-channel distributed MPPT architecture is critical in ensuring the best performance possible for vehicle-mounted PV systems, this thesis will continue to explore this concept and progress the MPPT design throughout the rest of this year.

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Appendix 1

Appendix 1 contains schematic printouts showcasing the current state of LV MPPT schematic capture. It also contains useful data sheet curves.

I2C Addresses:

CH1 - Input - 1000000
 CH1 - Output - 1000001
 CH2 - Input - 1000010
 CH2 - Output - 1000011
 CH3 - Input - 1000100
 CH3 - Output - 1000101
 CH4 - Input - 1000110
 CH4 - Output - 1000111
 CH5 - Input - 1001000
 CH5 - Output - 1001001
 CH6 - Input - 1001010
 CH6 - Output - 1001011

Figure 1: Power sensor I2C address

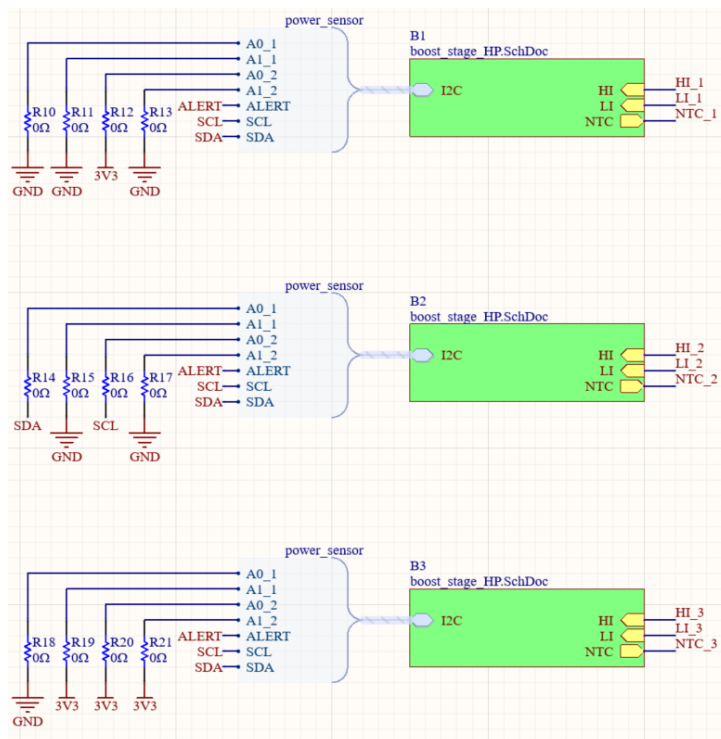


Figure 2: Top-level schematic view of channel 1, 2, and 3

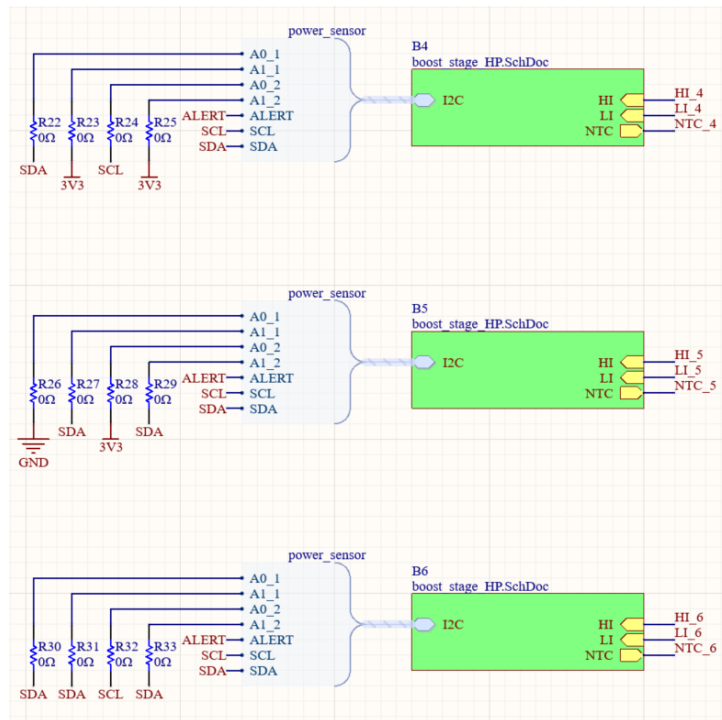


Figure 3: Top-level schematic view of channel 4, 5, and 6

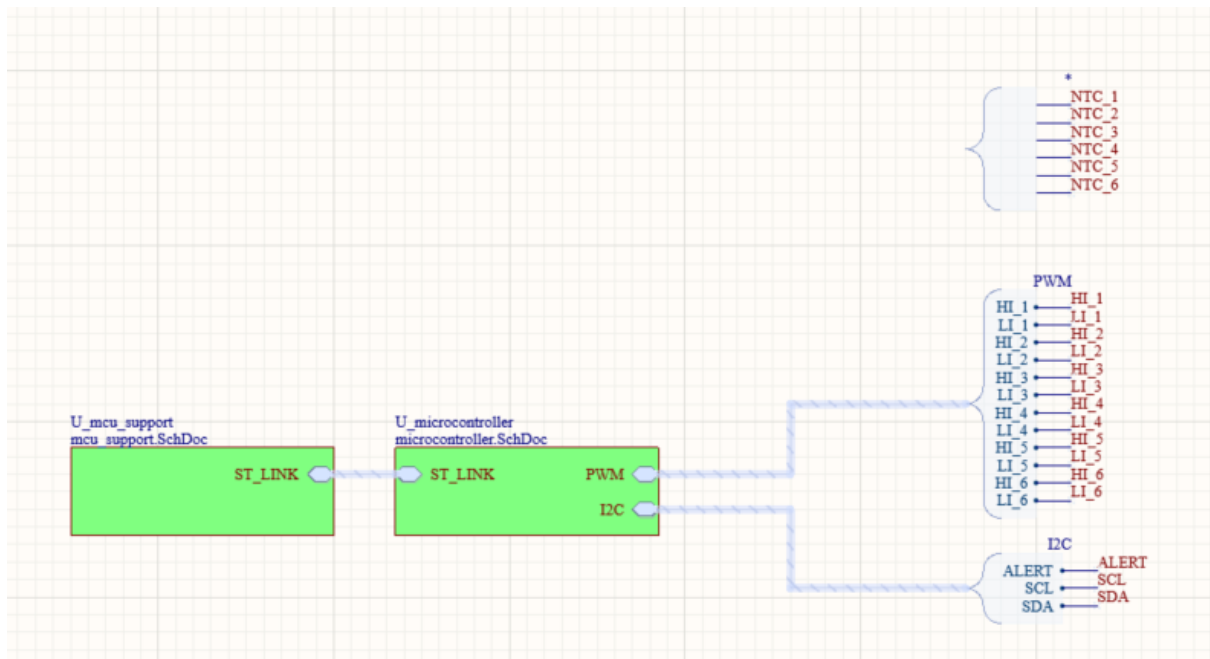


Figure 4: Top-level schematic view of MCU and MCU support blocks

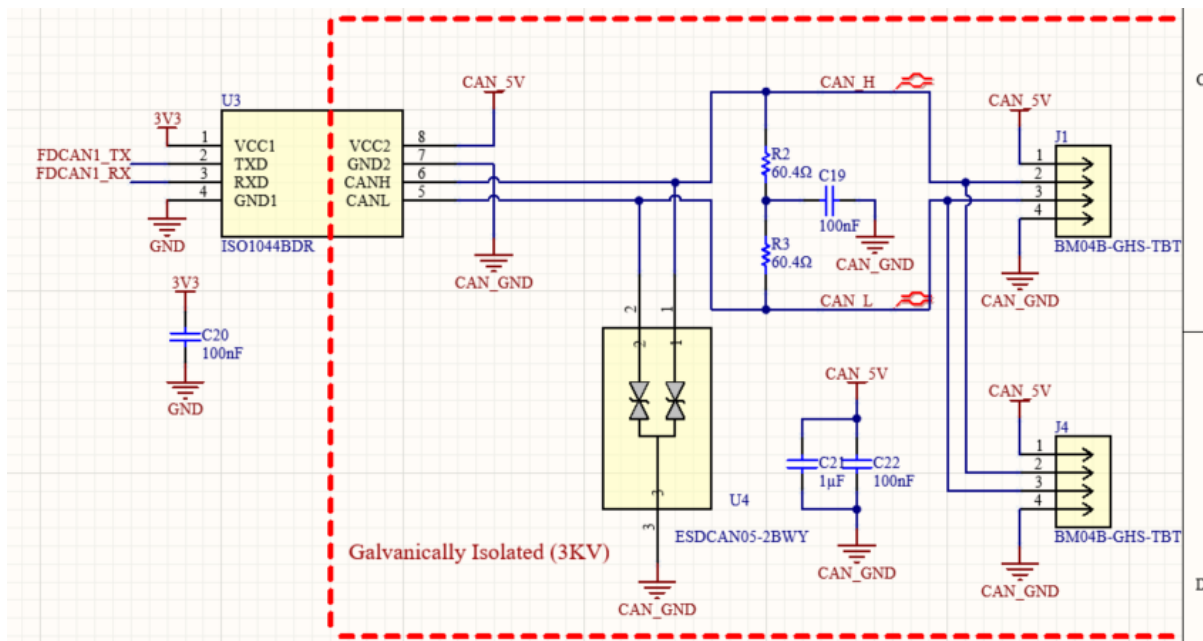


Figure 5: FD-CAN isolated transceiver circuit

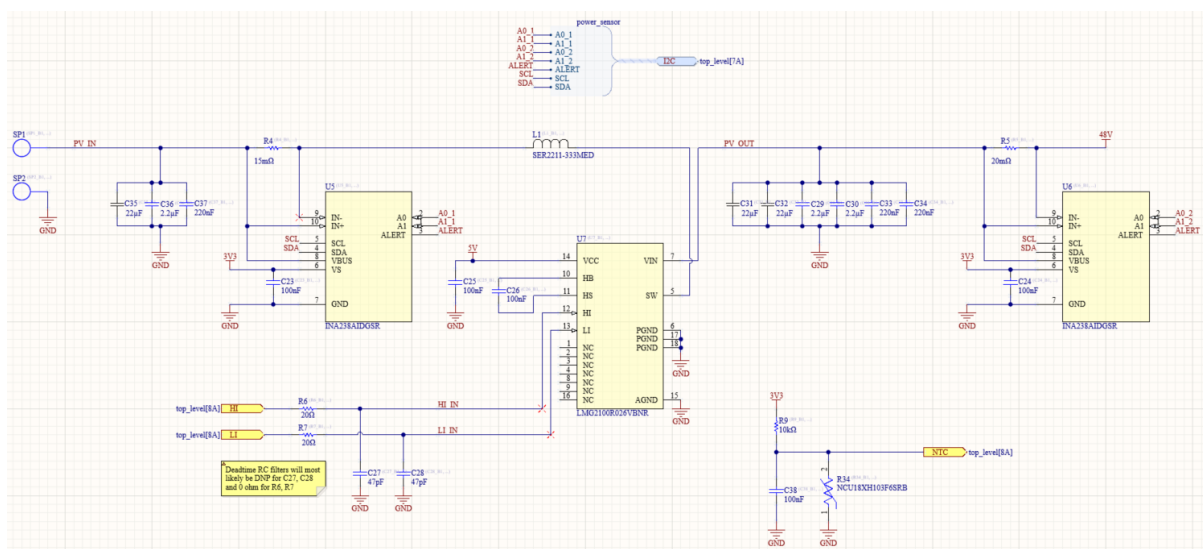
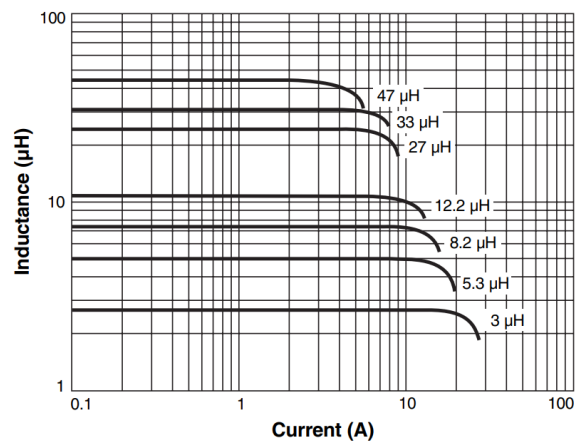


Figure 6: LV MPPT boost converter schematic

Typical L vs Current



Typical L vs Frequency

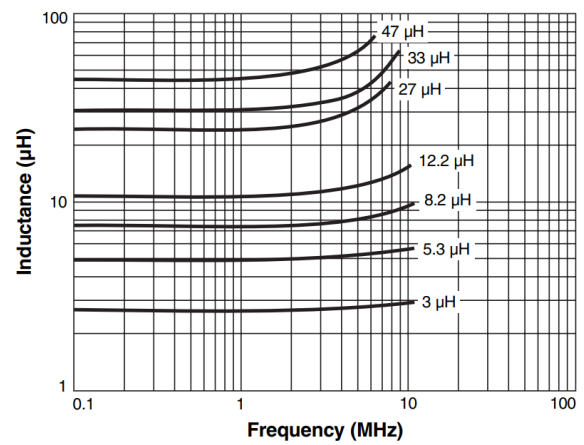


Figure 7: SER2211 series inductor curves

Appendix 2

HS017

HS Risk management form

For additional information refer to HS329 [Risk Management Procedure](#)

Faculty/Division: Electrical Engineering			School/Unit: School of Electrical and Telecommunicates Engineering						
Document number	1	Initial Issue date	23/4/26	Current version	1	Current Version Issue date	23/4/26	Next review date	TBD

Risk management name Undergraduate Thesis Risk Assessment - MPPT

Form completed by	Angus Macdonald	Signature	Date 23/4/26
Responsible supervisor/ authorising officer	John Fletcher	Signature	Date

Identify the activity and the location of the activity Description of activity Development of a multi-channel, dual-output, MPPT system for Sunswift Racing Description of location Electrical Engineering building and Sunswift Racing workshop		Identify who may be at risk from the activity: This may include fellow workers, visitors, contractors and the public. The types of people may affect the risk controls needed and the location may affect the number of people at risk Persons at risk Myself How they were consulted on the risk N/A
---	--	---

List legislation, standards, codes of practice, manufacturer's guidance etc used to determine control measures necessary

Work Health and Safety Act 2011
 Work Health and Safety Regulation 2011

HS Risk Management Form
 Version 3.5 23rd April 2013

Page 1 of 4

Identify hazards and control the risks.

1. An activity may be divided into tasks. For each task identify the hazards and associated risks. Also list the possible scenarios which could sooner or later cause harm.
2. Determine controls necessary based on legislation, codes of practice, Australian standards, manufacturer's instructions etc.
3. List existing risk controls and any additional controls that need to be implemented
4. Rate the risk once all controls are in place using the matrix in HS329 Risk Management Procedure

SHADED GREY AREAS

If you need to determine whether it's reasonably practicable to implement a control, based on the risk complete the shaded grey columns

Feel free to resize the boxes to suit your situation/the amount of text you need to use

Task/ Scenario	Hazard	Associated harm	Existing controls	Any additional controls required?	Risk Rating			Cost of controls (in terms of time, effort, money)	Is this reasonably practicable Y/N
					C	L	R		
Testing LV power electronic components	Capacitor burst, hot components, exposed LV, interacting with a battery	Eye damage, burns, very mild shocks, battery over-charging and catching on fire	• Wearing safety glasses, staying aware of hot components and avoiding touching system while active, staying aware of live board and do not touch while active, battery system has a BMS as well as MPPT will have voltage protection so a two stage system. In the event there is issues, sand is available in the workshop, and there is safe storage area for the battery when not in use.	No	2	D	L	N/A	N/A
Testing HV power electronic components	Capacitor burst, hot components, exposed HV, interacting with battery	Eye damage, burns, dangerous shocks, battery over-charging	• Wearing safety glasses, staying aware of hot components and avoid touching system while active, not touching HV board when active, wearing HV insulated gloves and using HV insulated tools that are now available in sunswift workshop, two factor protection to avoid battery overcharging with BMS protection and DUT protection. Testing HV component at lowest possible output voltage	HV testing may require additional training. Several Sunswift members have recently upskilled in this area, so may be able to test the device under my guidance.	4	D	M	If upskilling is required, considerable time and financial investment.	Yes
Working at a computer	Bad posture, looking at screen for too long	Eye strain and back strain	• Take regular breaks when working at a computer for extended periods of time	No	1	C	L	N/A	N/A
Soldering and assembling PCB's and enclosures	Inhaling fumes, lead solder, hot iron.	Flux irritation, lead ingestion, burns	• All UNSW soldering benches have extractors, use lead free wipers, maintain caution with hot iron. I have advanced solder badge so am certified for soldering on campus.	No	2	D	L	N/A	N/A

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RISK RATING METHODOLOGY AND MATRIX																																															
Consider the Consequences Consider: What type of harm could occur (minor, serious, death)? Is there anything that will influence the severity (e.g. proximity to hazard, person involved in task etc.). How many people are exposed to the hazard? Could one failure lead to other failures? Could a small event escalate?	Consider the Likelihood Consider: How often is the task done? Has an accident happened before (here or at another workplace)? How long are people exposed? How effective are the control measures? Does the environment affect it (e.g. lighting/temperature/pace)? What are people's behaviours (e.g. stress, panic, deadlines) What people are exposed (e.g. disabled, young workers etc.)?	Calculate the Risk 1. Take the consequences rating and select the correct column 2. Take the likelihood rating and select the correct row 3. Select the risk rating where the two ratings cross on the matrix below. VH = Very high, H = High, M = Medium, L = Low																																													
5. Severe: death or permanent disability to one or more persons 4. Major: hospital admission required 3. Moderate: medical treatment required 2. Minor: first aid required 1. Insignificant: injuries not requiring first aid	A. Almost certain: expected to occur in most circumstances B. Likely: will probably occur in most circumstances C. Possible: might occur occasionally D. Unlikely: could happen at some time E. Rare: may happen only in exceptional circumstances	<table border="1"> <thead> <tr> <th colspan="2"></th> <th colspan="5">CONSEQUENCES</th> </tr> <tr> <th colspan="2"></th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> </tr> </thead> <tbody> <tr> <th rowspan="5">LIKELIHOOD</th> <th>A</th> <td>M</td> <td>H</td> <td>H</td> <td>VH</td> <td>VH</td> </tr> <tr> <th>B</th> <td>M</td> <td>M</td> <td>H</td> <td>H</td> <td>VH</td> </tr> <tr> <th>C</th> <td>L</td> <td>M</td> <td>H</td> <td>H</td> <td>VH</td> </tr> <tr> <th>D</th> <td>L</td> <td>L</td> <td>M</td> <td>M</td> <td>H</td> </tr> <tr> <th>E</th> <td>L</td> <td>L</td> <td>M</td> <td>M</td> <td>M</td> </tr> </tbody> </table>			CONSEQUENCES							1	2	3	4	5	LIKELIHOOD	A	M	H	H	VH	VH	B	M	M	H	H	VH	C	L	M	H	H	VH	D	L	L	M	M	H	E	L	L	M	M	M
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	D	L	L	M	M	H																																									
	E	L	L	M	M	M																																									

Risk level	Required action
Very high	Act immediately: The proposed task or process activity must not proceed. Steps must be taken to lower the risk level to as low as reasonably practicable using the hierarchy of risk controls
High	Act today: The proposed activity can only proceed, provided that: (i) the risk level has been reduced to as low as reasonably practicable using the hierarchy of risk controls and (ii) the risk controls must include those identified in legislation, Australian Standards, Codes of Practice etc. and (iii) the document has been reviewed and approved by the Supervisor and (iv) a Safe Working Procedure or Safe Work Method has been prepared and (v) the supervisor must review and document the effectiveness of the implemented risk controls
Medium	Act this week: The proposed task or process can proceed, provided that: (i) the risk level has been reduced to as low as reasonably practicable using the hierarchy of controls and (ii) the document has been reviewed and approved by the Supervisor and (iii) a Safe Working Procedure or Safe Work Method has been prepared.
Low	Act this month: Managed by local documented routine procedures which must include application of the hierarchy of controls.

List emergency procedures and controls List emergency controls for how to deal with fires, spills or exposure to hazardous substances and/or emergency shutdown procedures Battery Fire - Ensure if any work is being done with batteries, the room is well-ventilated. Ensure sand is nearby to extinguish fire.
--

Implementation	Additional control measures needed:	Resources required	Responsible person	Date of implementation
	N/A	N/A	N/A	N/A

REVIEW Scheduled review date: Are all control measures in place? Are controls eliminating or minimising the risk? Are there any new problems with the risk? Review by: (name) Review date:			
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Acknowledgement of Understanding All persons performing these tasks must sign that they have read and understood the risk management (as described in HS329 Risk Management Procedure). Note: for activities which are low risk or include a large group of people (e.g. open days, BBQ's, student classes etc), only the persons undertaking the key activities need to sign below. For all others involved in such activities, the information can be covered by other methods including for example a safety briefing, induction, and/or safety information sheet (ensure the method of communicating this information is specified here)		
Risk management name and version number: Undergraduate Thesis Risk Assessment - MPPT - V1		
I have read and understand this risk management form		
Name	Signature	Date
Angus Macdonald		23/4/2026